



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

- i. How similar ions can be separated from each other with help of IEC?
- ii. What is hydrodynamic injection in capillary electrophoresis system?
- iii. What advantage does nitrous oxide acetylene flame offer for AAS?
- iv. What is chemical interference? Give example.
- v. What do you mean by the term "percent extraction"? Explain.
- vi. Why gel chromatography is also known as size exclusion chromatography?
- vii. What are the characteristics of anion exchange resin?
- viii. Define electro osmotic flow with its mathematical formula?
- ix. What is the basic difference between moving boundary and capillary zone electrophoresis?
- x. How is extraction efficiency defined?
- xi. How does electrophoresis help in gene and genome analysis?
- xii. What is meant by batch extraction?
- xiii. How is Hollow Cathode Lamp different than the lamp used in UV-Visible spectroscopy?
- xiv. Why solid phase extraction is an advanced technique for metal extraction?
- xv. How distribution ratio affects distribution law?

Q.2. Answer the following questions.

(6x5=30)

- a) What are ion exchange resins? What factors affect ion exchange equilibria?
- b) Explain the counter current process in solid phase extraction.
- c) What is solid phase extraction? Explain its working.
- d) Discuss different events occurring in flame in flame emission spectroscopy with the help of a diagram?
- e) Explain different gels used in electrophoresis.
- f) Explain zeta potential in detail on basis of electroosmotic flow.

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Short Answers

(i)

Extraction efficiency:

The extraction efficiency is the percentage of solute that moves from one phase to the other.

It is not more than 100%.

(ii)

Hydrodynamic injection:

Hydrodynamic injection is used to introduce liquid sample into capillary for separation & analysis.

Unlike electrokinetic injection in which sample is introduced into capillary using applied voltage, Hydrodynamic injection uses pressure or vacuum to physically push sample into capillary.

(iii)

Similar ion separation by IEC

Similar ions are separated in ion exchange

chromatography based on their charge density & affinity to resin. Ions with higher charge or small size bind more strongly to the ion exchange resin and elute later than those with weaker interactions

(iv)

Nitrous-oxide acetylene flame:

The nitrous oxide acetylene flame is both hot and reducing. A characteristic red interconal zone is obtained under slightly fuel-rich is due to the emission by cyanogen radical. This radical is a very efficient scavenger for oxygen.

(v)

Extraction efficiency:

Extraction efficiency is percentage of a substance

successfully removed from
one phase to another
— during an extraction
process.

$$\text{Extraction efficiency (\%)} = \frac{\text{Amount extracted} \times 100}{\text{totally amount initially present.}}$$

(vi)

How distribution ratio affects distribution law?

Distribution ratio affects the distribution law by determining whether the solute exists in the same form in both phases. If it doesn't due to (due to ionization or complexation), the distribution law must be modified, as the ratio no longer reflects true equilibrium constant.

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E-P helps in gene & genome analysis :

Electrophoresis help in gene & genome analysis by separating DNA fragments based on size, allowing researchers to identify, compare & analyze genetics material for mutations, gene mapping and sequencing.

(viii)

Electro-osmotic flow :

It simply refers to the movement of material through a porous material due to applied electric field.

Formula:

$$U_{EOF} = \frac{-\epsilon \zeta_0 E}{\mu}$$

ζ_0 = zeta potential

E = electric field

μ = dynamic viscosity

ϵ = relative permittivity.

This phenomenon is useful for chemical separation techniques and buffered solutions. — (ix) —

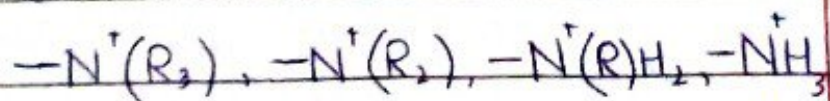
Batch extraction:

Batch extraction is a solvent extraction technique where solute is separated from the solution in discrete steps by mixing it with solvent in separate batches rather than continuously.

— (x) —

Characteristics of anion exchange resin:

Anion exchange resin is a high molecular weight cross linked polymer having groups :



as integral part of resin and equivalent amount of anion such as Cl^- , SO_4^{2-} & OH^- .

• Thus anion exchange

Resins contain a polymeric cation to which active anion are attached.

- They can be represented as $(ReB^+)A^-$.

Re is polymeric resin, B^+ is cation & A^- is mobile anion

- When anion exchange resins are placed in solution, the mobile or active anion of resin is exchanged by anion of solution.



(xi)

Hollow cathode lamp is diff from UV-vis lamp

A Hollow Cathode Lamp is used in AAS

(atomic absorption spectroscopy).

It emits sharp element-specific lines because it contains a cathode made up of element to be analyzed. In contrast the lamp used in UV-vis spectroscopy (deuterium or tungsten lamp) emits a broad spectrum of light and is not element-specific.

Gel chromatography also called SEC. Why?

Because size exclusion chromatography (SEC) separates ions on the basis of their size, filtering them through a gel. Gel contains (specific) spherical beads containing pores of a specific size distribution.

Separation occurs when molecules of different size are included or excluded from pores within the matrix.

Gel chromatography also separates different size of molecules.

(xiii)

Difference b/w Capillary E-P and Moving boundary

It uses a thin capillary tube filled with electrolyte to separate ions based on their charge to size

It involves separation of charged particles in a free solution without supporting media, observed as boundaries moving in

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ration under high voltage offers high resolution & automation.

a U-tube.

It is older & less precise method.

(xiv)

Chemical interference

-nce :

It refers to any chemical reaction or interaction in flame or furnace that reduces the amount of free atoms in analyte thus affecting absorption & accuracy.

e.g.:

In calcium analysis phosphate ions can form stable calcium phosphate, preventing free calcium atoms for from forming. This reduces calcium absorption signals.

Solid phase extraction is advanced technique for metal extraction.

Reasons:

Higher selectivity:

SPE uses solid adsorbent material that selectively bind to specific metal ions, reducing interference & improving accuracy.

Greater Sensitivity:

SPE can detected lower concentration of metals due to preconcentration effect of solid phase.

Reduce solvent SPE minimizes uses of organic solvents making it more environment friendly & cost effective. option.

Improved safety:

SPE reduced risk of exposure to toxic solvents & metals.

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